Fundusze Europejskie Wiedza Edukacja Rozwój
Politechnika Wrocławska
Unia Europejska Europejski Fundusz Społeczny

„ZPR PWr – Zintegrowany Program Rozwoju Politechniki Wrocławskiej”

Supporting tools
Maya, 3DS Max, Blender

Marek Kopel

wideo z wykładu 2022:
https://pwr-edu.zoom.us/rec/share/aO3GrwuHV-XwZlgYReA_2h2JkNTVfQ6v3bcN3N0IBC60HUiX9UdZX1TMlwir5A5N.9iofV2ryGKF9L2pa

dr inż. Marek Kopel

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eportal.pwr.edu.pl

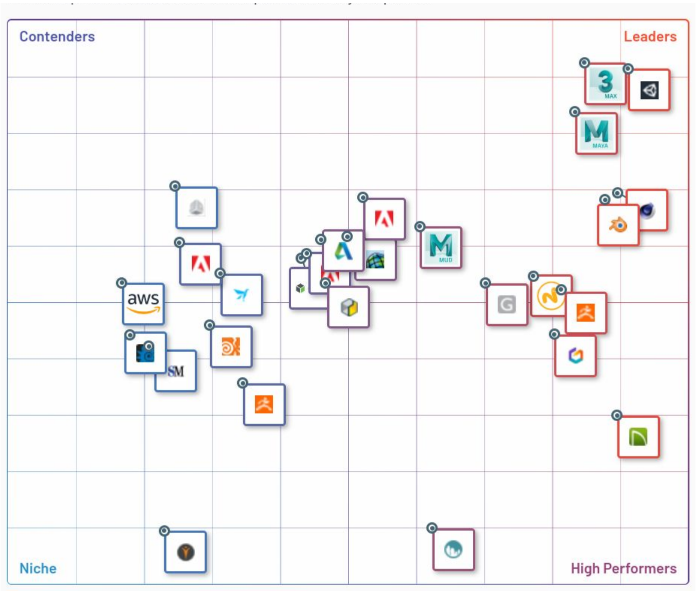
“Programowanie gier”

Info o ułatwianiu dostępności: Osoby, które ze względu na stan zdrowia, niepełnosprawność lub inne obiektywne przesłanki mogą mieć szczególne potrzeby związane ze sposobem realizacji zajęć, zaliczenia bądź przygotowaniem materiałów proszone są o zgłoszenie się na konsultacje lub po zajęciach czy napisanie e-maila w tej sprawie. Będę starał się, aby na moich zajęciach każdy miał **równe prawo do zdobycia wiedzy i rozliczenia się z niej**

Lecture

1. **Silniki gier. Wprowadzenie do Unity. Pierwsza gra 2D.**
2. **Mechanika 2D, animacje.**
3. **Oświetlenie, tekstury, materiały. Pierwsza gra 3D.**
4. Narzędzia wspierające, np. Blender.
5. **Zapis/odczyt danych, komunikacja sieciowa.**
6. **Modelowanie terenu. Generowanie terenu.**
7. **Sztuczna inteligencja w grach.**
8. Virtual Reality, wsparcie VR w Unity.
9. **Optymalizacje w Unity.**
10. Historia i klasyfikacja gier.
11. Prototypowanie gier. GDD.
12. Projektowanie poziomów gier.
13. Projektowanie gier dla różnych platform.
14. Testowanie gier. Test.
15. Test.

3D modeling tools comparisons



https://www.g2.com/categories/3d-modeling

- grid view

https://en.wikipedia.org/wiki/Comparison_of_3D_computer_graphics_software

https://en.wikipedia.org/wiki/List_of_3D_modeling_software

- complete list

https://www.creativebloq.com/features/best-3d-modelling-software

- best of (2025)

https://all3dp.com/1/best-free-3d-modeling-software-3d-cad-3d-design-software/

- table w/ skill level and supported formats ?!?

Autodesk Maya

An industrial-strength powerhouse, with a price to match

Cost: £222/month, £1,782/year | Pricing model: Subscription OS: Win, Mac, Lin

- Incredibly powerful
- Comprehensive CG toolset
- Expensive and difficult to learn
- Overkill for many 3D artists

Ask any 3D artist to name the best 3D modelling software, and most will choose Autodesk Maya. **Largely seen as the industry standard for CG**, Autodesk Maya boasts an unrivalled range of tools and features. This **hugely extensible** app isn't for the faint-hearted: its toolset is hugely complex and takes time to learn. However, if you're aiming to get a job in the **animation or VFX industries**, you'd be wise to use the same software that the likes of ILM, Pixar, DNEG and Framestore use.

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from: https://www.quora.com/What-games-or-films-have-been-made-using-Maya

Autodesk Maya

> Maya has been used to create graphics for many cinematic films with Pixar's Renderman, including the Academy Award winners, *Monsters, Inc.*, *The Matrix*, *Spider-Man*, *The Girl with the Dragon Tattoo*, *Avatar*, *Finding Nemo*, *Up*, *Hugo*, *Rango*, and *Frozen*. It is also used to create visual effects for television programs, including *Game of Thrones*, *The Walking Dead*, *Once Upon a Time*, *Bones*, *Futurama*, *Boardwalk Empire* and *South Park*.
> Maya is involved in creating the visual effects for video games, including *Halo 4*.

Autodesk Maya cont.

An industrial-strength powerhouse, with a price to match

Maya is great at **modelling, texturing, lighting and rendering** – its vast feature set includes **particles, hair, solid body physics, cloth, fluid simulations and character animation**. There's a chance you may never touch some of its functionality, so you need to decide if it's actually overkill for your specific needs.

Type: Subscription **Price:** $245/month

When you ask any professional 3D modeller in the industry which program they use the most, Autodesk Maya is the most common answer, and for good reason. Most of the leading animation studios use it (Pixar included) due in part to the massive array of powerful tools offered within the package. The last few years have seen some especially amazing new features burst into the limelight such as truly **jaw-dropping live rendering**.

Autodesk 3Ds Max

The best 3D modelling software for Windows users

Cost: £222/mo, £1,782/yr | Pricing model: Subscription OS: Win

- Easier to learn than Maya
- Substantial feature set for modelling and VFX
- Windows only
- Recent updates have been underwhelming

3ds Max is Autodesk's PC-only 3D computer graphics program, used for **TV and feature film production and for architectural and product visualisation**. Like its sister software Maya, 3ds Max boasts a very robust toolset for 3D modelling, not to mention **fluid simulations, hair and fur, plus character rigging and animation**.

It uses both **direct manipulation and procedural modelling** techniques, and a huge **library of different modifiers** makes the modelling process **easier for new** or intermediate 3D artists.

3ds Max offers a professional toolset and, unsurprisingly, comes with a professional price tag. However, s**tudents can get the software for free** and a trial version is also available for 30 days.

Autodesk 3Ds Max cont.

The best 3D modelling software for Windows users

It is particularly popular among **video game developers, visual effects artists, and architectural visualization** studios. Sophisticated **particle and light simulation, cloth-simulation engine and its own scripting language (MAXScript)** are just a few key features besides its 3D modeling capabilities. If that is not sufficient for you, 3ds Max comes with a **plugin architecture** that is continuously fed by a vibrant community of developers. Up until the current version (2017) this 3D modeling software was split into a **version for engineers and designers (3ds Max Design)** and a **version for visual artists (3ds Max)**, which shared only the core functionalities.

In terms of 3D modeling itself, 3ds Max is capable of creating **parametric and organic objects with polygon, subdivision surface, and spline-based modeling** features. Features interesting (for designers in particular) are the NURBS-based modeling tools (comp. w/ mesh) in this 3D modeling software that allows for **both organic and mathematically precise meshes**. Among the more other techniques is the ability to **create models from point cloud data**. (Radiohead - “no camera” video)

3ds Max it is one the best 3D modeling software solutions out there. This being said, the animation and engineering features require long training in order to fully master them.

Autodesk 3Ds Max cont.

The best 3D modelling software for Windows users

Type: Subscription **Price:**$216/month to $1740/year

3Ds Max has been **around for a long time, as far as modeling software goes**. It predates almost every other current program by several years and has plenty of performance patches under its belt as a result. It’s **one of the most stable 3D modeling programs around (probably the most stable option on Windows, period)** and has a **gigantic library** available that provides access to countless functions that can **make the process of modelling less tedious**. Many of these **mods also make things easier for beginners**.

It has a higher price point but offers **free student licenses** and a trial that allows you access to all of the features the app has to offer for 30 days, which should provide further incentive to give it a go and wow your clients.

Blender

FLOSS - Free and open-source software

If you're serious about 3D but struggling to afford software, then you're in luck. Blender is a **free, open source** 3D content creation suite, available for **all major operating systems**.

Started by Blender Foundation founder Ton Roosendaal back in 2002, Blender is now **largest open source tool for 3D** creation. Its makers are constantly working on its development, but you can pretty much do anything 3D related with this software, from **modelling, texturing and animation to rendering and compositing**. It even includes tools for creating **hand drawn 2D animation**.

Blender is a intermediate-to-professional open-source and free 3D modeling software for creating animated films, visual effects, art, **interactive applications, video games** – and 3D printed models. Blender's dizzying array of features includes **3D modeling, UV unwrapping, texturing, raster graphics editing, rigging, and skinning, fluid and smoke simulation, particle simulation, soft body simulation, sculpting, animating, match moving, camera tracking, rendering, video editing, and compositing**. In addition, this free 3D modeling software features an **integrated game engine**.

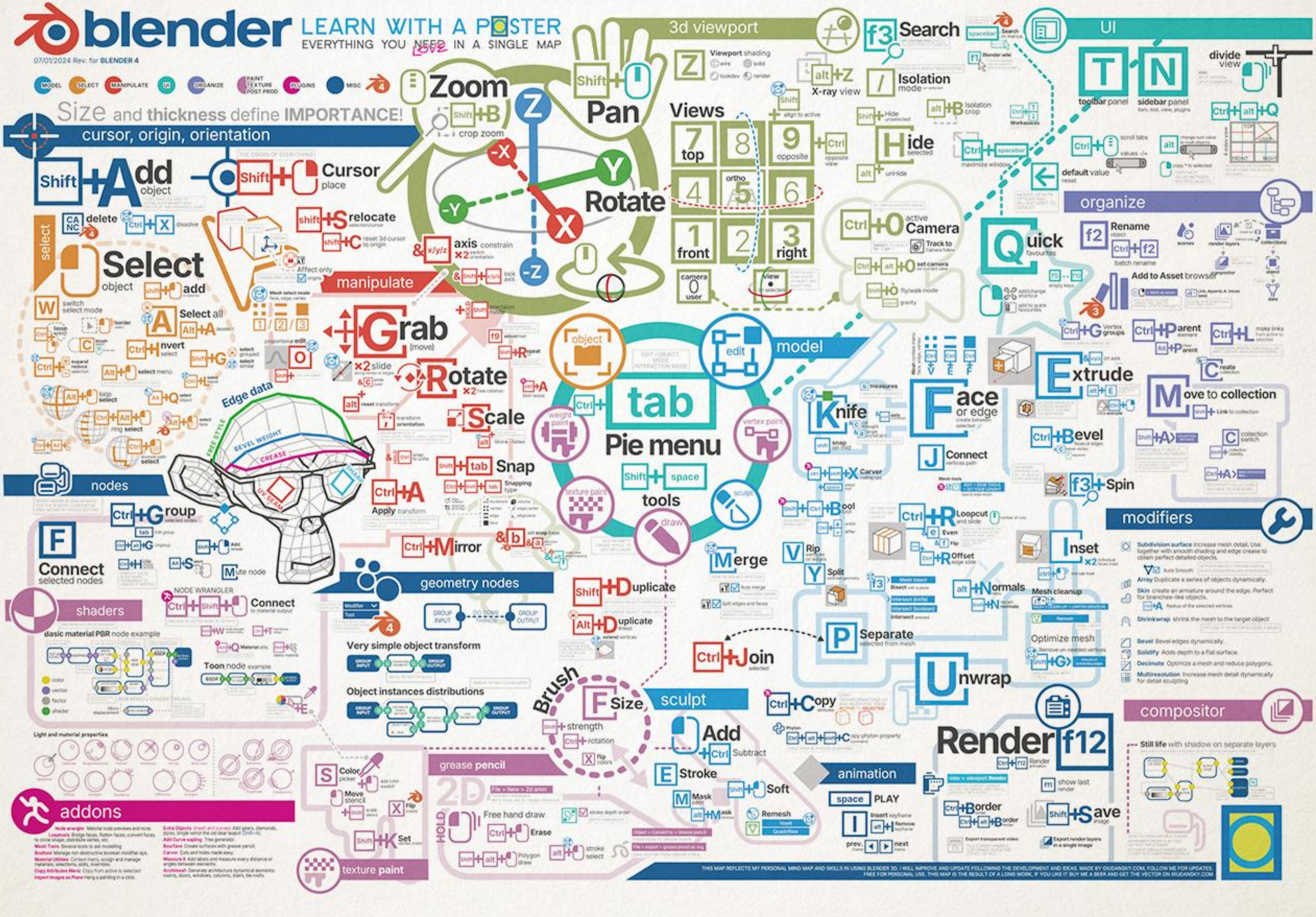
It is clear that such wealth of functionality comes at a price. Although this 3D modeling software is free, it is by no means easy to master.

Blender cont.
FLOSS - Free and open-source software

In terms of modeling, it is noteworthy that 3D design software Blender includes **sculpting capabilities similar to Mudbox or ZBrush**. This free 3D modeling software provides a number of tools and modifiers that ease the **creation of meshes intended for 3D printing – including a solution to repair meshes**.

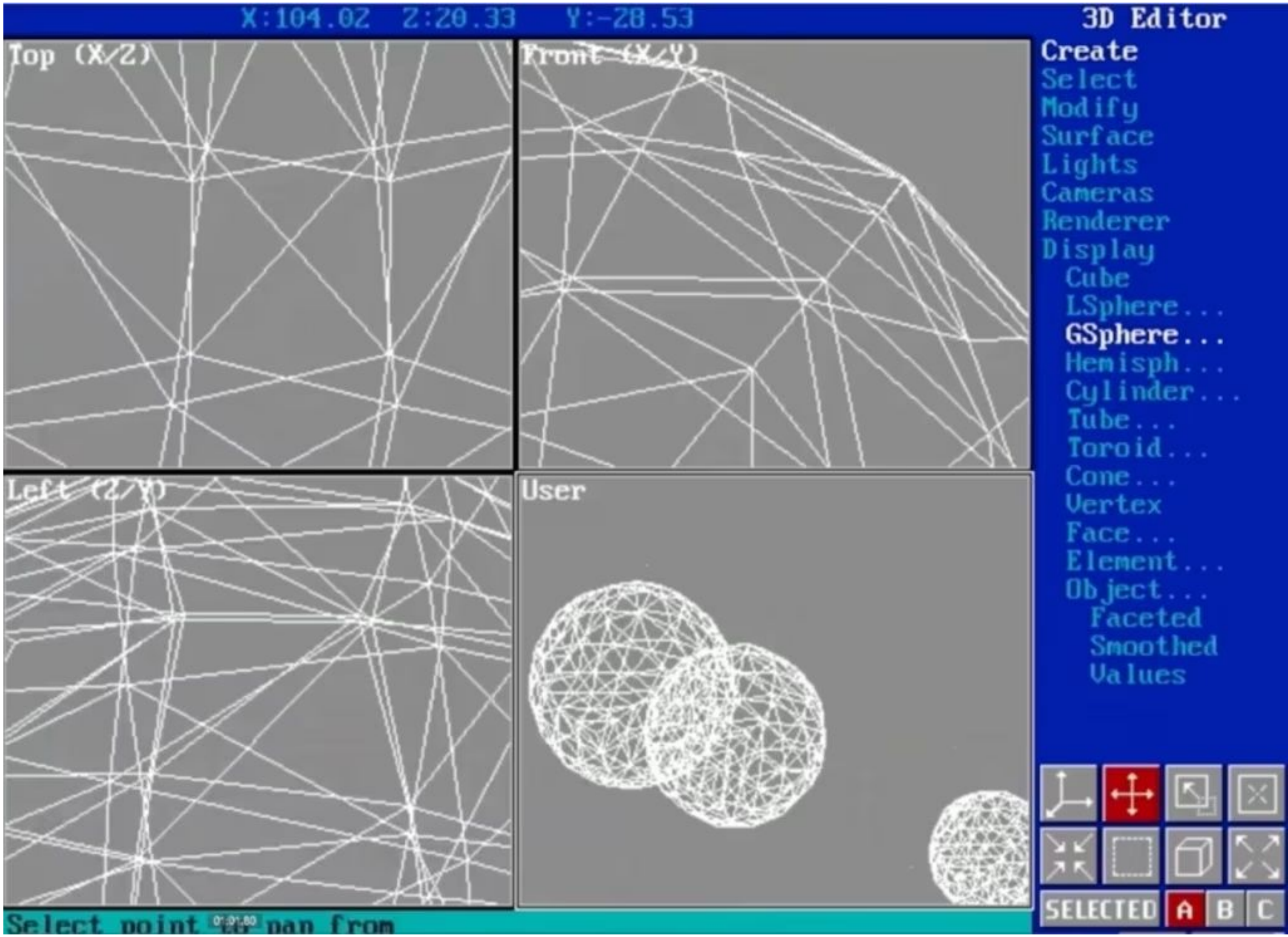
Blender is not suited for beginners, but if you are not turned off by a **steep learning curve** and search for one of the best free 3D modeling software that is a jack of all trades, you have found it.

As an added bonus, it's open-source! This means that there are constant improvements and available add-ons for new functionality are not only common, but always free of charge. This puts a lot of power in your hands. If you can't swing an expensive license for Maya, this is the next best thing.


blender
LEARN WITH A POSTER
EVERYTHING YOU NEED LOVE IN A SINGLE MAP
07/01/2024 Rev. for BLENDER 4
MODEL
SELECT
MANIPULATE
UI
ORGANIZE
PAINT TEXTURE POST PROD
PLUGINS
MISC
Size and thickness define IMPORTANCE!
cursor, origin, orientation
Shift + Add object
Shift + Cursor place
shift + S relocate
shift + C reset 3d cursor to origin
& x/y/z axis constrain
×2 switch orientation
Affect only
CA NC delete
Ctrl + X dissolve
select
Select object
shift + add
W switch select mode
A Select all
Alt + A deselect
Ctrl + I nvert select
Shift + G select grouped
L select
Ctrl + L
loop select
ring select
Edge data
BEVEL WEIGHT
CREASE
SHARP
UV SEAM
FREE STYLE
manipulate
Grab (move)
×2 slide
Rotate
×2 free rotation
alt reset transform
Scale
Shift + tab Snap
Snapping type
Ctrl + A Apply transform
Ctrl + Mirror
O proportional edit
nodes
Ctrl + Group selected nodes
tab Edit group
F Connect selected nodes
M ute node
NODE WRANGLER
Ctrl + Shift + Connect to material output
shaders
basic material PBR node example
Toon node example
color
vector
factor
shader
Light and material properties
addons
geometry nodes
GROUP INPUT
DO SOME THINGS
GROUP OUTPUT
Very simple object transform
Object instances distributions
texture paint
S Color picker
Move stencil
X Flip colors
Shift + K Set vertex color
grease pencil
2D
Free hand draw
Ctrl + Erase
Polygon draw
alt + stroke select
HOLD
3d viewport
Zoom
Shift + B crop zoom
Pan
Shift +
Rotate
Z
Y
X
-X
-Y
-Z
Views
7 top
8
9 opposite
4
ortho 5
6
1 front
2
3 right
camera 0 user
view selected
Ctrl +
Z Viewport shading
wire
solid
lookdev
render
alt + Z X-ray view
/ Isolation mode
alt + B Isolation crop
H ide selected
Shift + H Hide unselected
alt + H unHide
Ctrl + 0 active Camera
Track to
Ctrl + alt + 0 set camera
Shift + fly/walk mode
gravity
f3 Search
spacebar
f1 Blender wiki
UI
T toolbar panel
N sidebar panel
divide view
Ctrl + alt + Q
Ctrl + spacebar maximize window
Ctrl + scroll tabs
alt + default value reset
organize
Q uick favourites
f2 Rename object
Ctrl + f2 batch rename
Add to Asset browser
Ctrl + G Vertex groups
Ctrl + P arent
Ctrl + L make links
C reate collection
M ove to collection
Shift + A
C collection switch
Ctrl + A
tab
object
edit
model
Ctrl +
Pie menu
Shift + space tools
weight paint
vertex paint
texture paint
sculpt
draw
Brush
F Size
Shift + strength
Ctrl + rotation
X flip colors
sculpt
Add
Ctrl Subtract
E Stroke
M Mask
Shift + Soft
Remesh
Knife
snap on mid
Shift + X Carver
Face or edge
J Connect vertices path
Ctrl + B evel
E xtrude
f3 + Spin
Ctrl + Bool
Ctrl + R Loopcut and slide
Inset
×2 individual faces inset
Offset edge slide
Even
Flip
V Rip
M erge
Y Split
Bisect with a plane
Intersect
alt + N ormals
Mesh cleanup
Optimize mesh
Shift + D uplicate
Alt + D uplicate linked
Ctrl + J oin selected
P Separate selected from mesh
U nwrap
Ctrl + C opy
animation
space PLAY
I Insert keyframe
alt + I Remove keyframe
prev. next frame
modifiers
Subdivision surface
Array
Skin
Shrinkwrap
Bevel
Solidify
Decimate
Multiresolution
Render f12
Ctrl + f12 Render animation
f11 show last render
Ctrl + B order
Ctrl + alt + B order
Shift + S ave image
compositor
Still life with shadow on separate layers
THIS MAP REFLECTS MY PERSONAL MIND MAP AND SKILLS IN USING BLENDER 3D. I WILL IMPROVE AND UPDATE FOLLOWING THE DEVELOPMENT AND IDEAS. MADE BY GIUDANSKY.COM, FOLLOW ME FOR UPDATES.

Brief history: 3DS Max

https://beforesandafters.com/2020/06/04/a-visual-history-of-3ds-max/



Making 3D look analogue

Paperman

https://youtu.be/TZJLtujW6FY

Fake camera in Toy Story 4

https://youtu.be/AcZ2OY5-TeM

Check Engine Light exhibition by Joshua Vides | Joshua Vides, a Southern California artist, is known for his "Reality to Idea" (RTI) technique, which turns real objects into black-and-white,... | By Molten Immersive Art | Facebook

HERO – **Blender** Grease Pencil Showcase (watch **closing credits**)

https://www.youtube.com/watch?v=pKmSdY56VtY

#hellboy rig. Made with **Moho**!

https://www.facebook.com/TheMikeRoberts/videos/566598787128453/

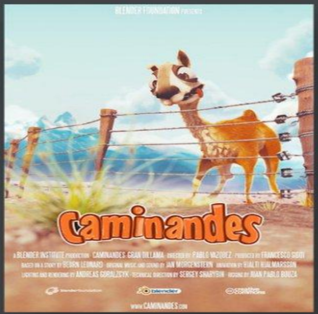
ALL BLENDER OPEN MOVIES TILL DATE

2006

2008

2010

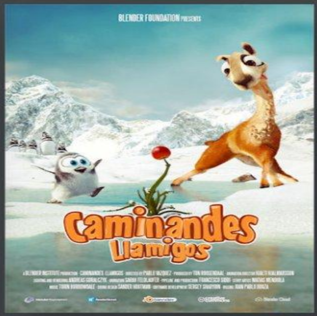
2012



2013

2015

2015



2016

2017

2017

2018

2019

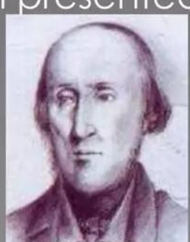
2020

2021

Timeline Of Animation

1824

Peter Mark Roget researched in University that you can with still images you can create the appearance of still images and published a book which presented that idea



1832

Joseph Plateau developed the phenakistiscope

1914

Windsor McCay produces perhaps the first popular animation 'Gertie The Dinosaur

1917

John bray patent rotoscoping

1920

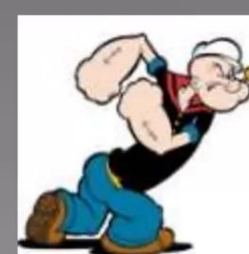
Otto Mesmer creates 'Felix The Cat'

1928

Walk Disney release 'Steamboat Willie'

1930

Fleischer studios create 'Betty Boop' and 'Popeye'



1937

'Snow White' is released at a cost of $1.5M

1993

'Jurassic Park' use of CG (computer generated) for realistic living creatures

1995

'Toy Story' first full-length 3D CG (computer generated) feature film

Principles of animation

tylko **ZAJAWKA**
- całość na
Gry Komputerowe
II stopień,
spec. PSI

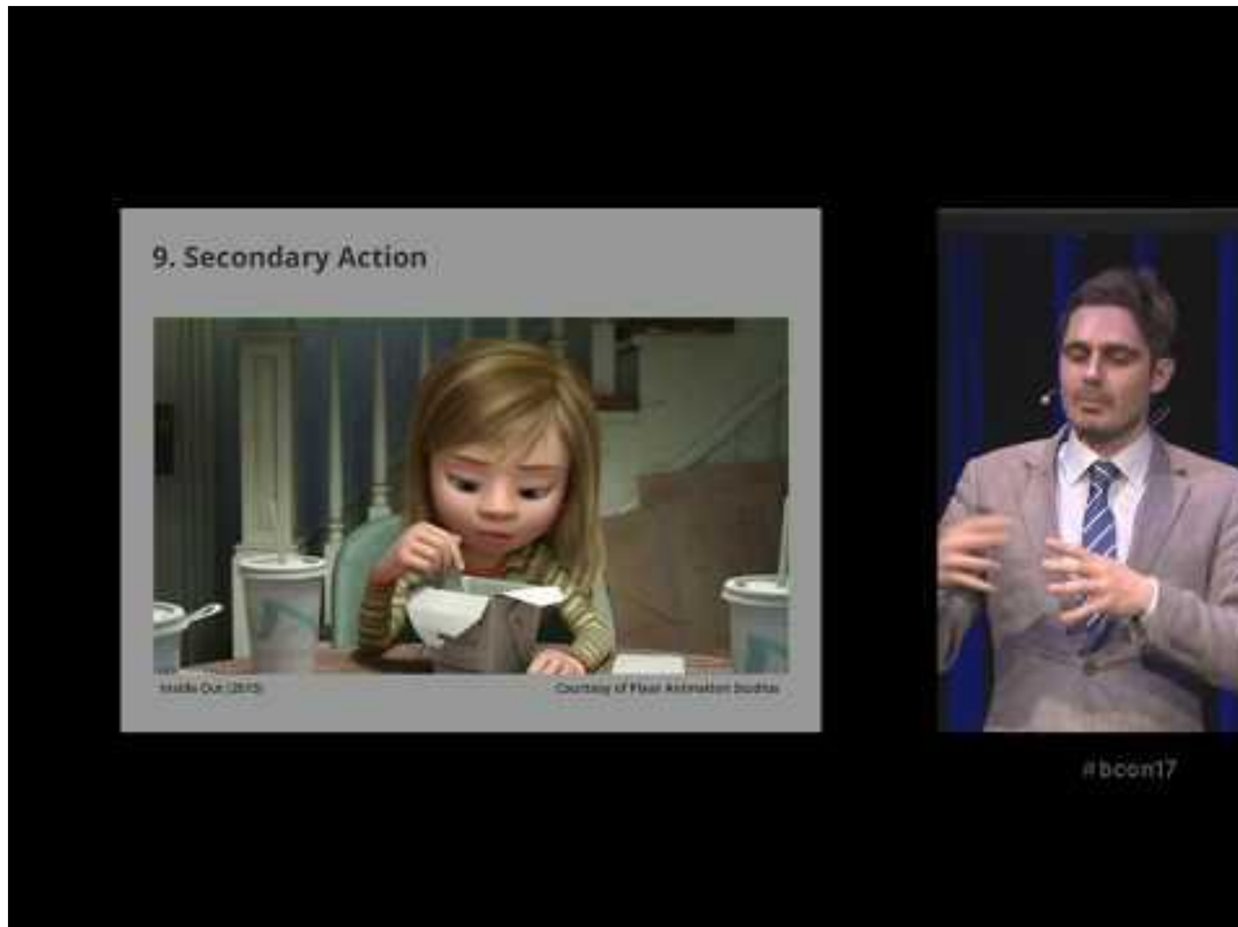
https://en.wikipedia.org/wiki/Disney%27s_Nine_Old_Men#Leg

12 basic principles of animation:

1. Squash and stretch
2. Anticipation
3. Staging
4. Straight Ahead Action and Pose to Pose
5. Follow Through and Overlapping Action
6. Slow In and Slow Out
7. Arcs
8. Secondary Action
9. Timing
10. Exaggeration
11. Solid Drawing
12. Appeal

Principles of animation - cont.

1. Timing & Rhythm
2. Spacing
3. Squash & Stretch
4. Arcs
5. Anticipation
6. Drag & Follow Through
7. Asymmetry
8. Overlapping Action
9. Secondary Action
10. Exaggeration
11. Consistency
12. Staging & Clarity
13. Readability & Focus



Rigging

RAIN v2.0

https://cloud.blender.org/p/characters/5f04a68bb5f1a2612f7b29da

How long a Pixar movie renders?

https://www.facebook.com/photo.php?fbid=10152958771619725&set=pb.530059724.-2207520000.&type=3

[All 54 Modifiers in Blender Explained in 10 Minutes]